

HONG WOON YUN

+82-10-2374-1734 ◇ yhw4241@gmail.com

OBJECTIVE

To contribute to the successful execution of the AI Capstone Design project

PERSONAL PROFILE

Dedicated researcher specializing in the fabrication and characterization of neuromorphic semiconductor devices based on advanced materials, including 2D materials (TMDs) and oxide semiconductors. My core expertise lies in developing floating-gate-based memristors to implement high-performance synaptic and neuronal functions. I am committed to advancing large-scale integrated neuromorphic systems that bridge the gap between device-level innovation and hardware-level neural network implementation

EDUCATION

Ph.D in AI System Engineering September 2024 - Present
Sungkyunkwan University (SKKU), Suwon, South Korea.
Overall GPA: 4.36 / 4.5 (27 credits completed)

Master of Science in AI System Engineering September 2022 - August 2024
Sungkyunkwan University (SKKU), Suwon, South Korea.
GPA: 4.06 / 4.5
Thesis: [Two-terminal Au Nanoparticle Floating-gate Memristor for Neuromorphic Computing]

Bachelor of Engineering March 2016 - August 2022
Sungkyunkwan University (SKKU), Suwon, South Korea.
Dual Major: Systems Management Engineering & Electronic and Electrical Engineering
Overall GPA: 3.69 / 4.5
Major GPA (Electronic and Electrical Engineering): 4.12 / 4.5

PROJECTS

Van der Waals Heterostructure-Based Multi-terminal Floating-Gate Memristors and Unsupervised Learning Neuromorphic Systems March 2022 - February 2026

Development of Core Design and Fabrication Technologies for Low-Power, High-Speed Floating-Gate (FG) Memristors March 2022 - January 2024

High Energy-Efficient Integrated Neuromorphic Vision System Based on 2D Semiconductor Materials for Artificial Retina and Visual Cortex Applications July 2024 - Present

Development of Energy-Efficient 2D Neuromorphic Devices and Circuits for Edge AI Computing - July 2024 - Present

Implementation of a CNN Pooling Layer Using Spatiotemporal Summation in Multi-terminal Short-term Floating-gate Devices March 2025 - Present

CAPSTONE DESIGN

Project title : Multi-terminal Short-term Floating-gate Memristor for Implementing Convolutional Neural Network

Keywords : Neuromorphic, Reservoir computing, Convolutional Neural Network, Floating-gate Memory

Project achievements : Patent application 1, Software copyright registration 1